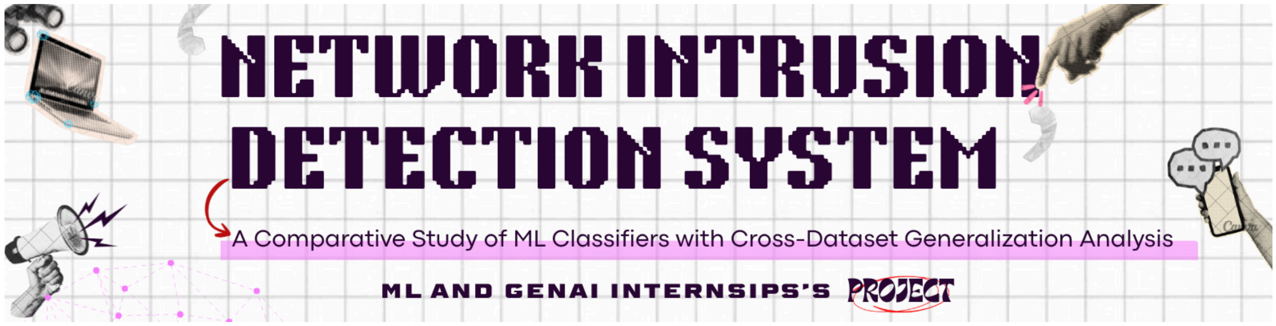

🔗kaggle link-https://www.kaggle.com/code/sneharani20/network-intrusion-detection-system

Members and their Enrollment Nos

> **1.Sneha Rani-12601182025**
>
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>
> **2.Sneha Verma-06401192025**

Project Title

`Cross-Dataset Generalization Analysis of Machine Learning Classifiers for Network Intrusion Detection.`

Problem Statement

Traditional Network Intrusion Detection Systems (NIDS) are typically evaluated on a single benchmark dataset, which can produce misleadingly high accuracy that doesn't hold up on real-world or differently-sourced traffic. This project trains and compares multiple machine learning classifiers on **NSL-KDD, then explicitly tests how well they generalize to a dataset of a different vintage (KDD Cup 1999)** without retraining, to measure the true generalization gap rather than reporting single-dataset accuracy alone.

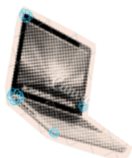
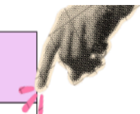
Objective of the Project

1. Preprocess and encode the NSL-KDD dataset correctly (no data leakage)
2. Train and compare 10 ML classifiers on NSL-KDD using accuracy, precision, recall, and F1-score
3. Test the same trained models on KDD Cup 1999 without retraining, to measure cross-dataset generalization
4. Identify which classifier is most **accurate** vs which is most **robust/generalizable** — these turn out to be different models
5. Additionally benchmark the same algorithms independently on a modern dataset (CICIDS2017) to see whether relative rankings hold across a different feature representation.

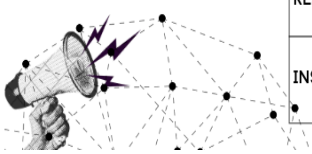
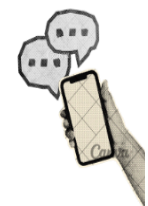
Dataset Name (with source)

- NSL-KDD
- KDD Cup 1999
- CICIDS2017

USP/Research Gap

Aspect	Typical Projects	Our Project
DATASETS	SINGLE (NSL-KDD)	THREE (NSL-KDD + KDD99 + CICIDS2017)
EVALUATION	TRAIN/TEST SAME DATASET	TRAIN ON NSL-KDD → TEST ON DIFFERENT DATASETS
RESEARCH QUESTION	"WHAT'S THE BEST MODEL?"	"WHICH MODEL GENERALIZES BEST ACROSS DATASETS?"
INSIGHT	HIGH ACCURACY CLAIMS	GENERALIZATION GAP ANALYSIS – REVEALS WHICH MODELS ARE DATASET-DEPENDENT

Number of Samples and Features

Dataset	Sample	Feature	Notes
NSL-KDD Train	125,973	41	`difficulty` column dropped
NSL-KDD Test	22,544	41	Official benchmark split, includes attack types absent from training
KDD Cup 1999 (10% sample)	494,021	41	Same 41-feature schema as NSL-KDD
CICIDS2017 (10% sample used)	252,075	52	Different, flow-based feature schema — not directly comparable to NSL-KDD

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Data Preprocessing Performed

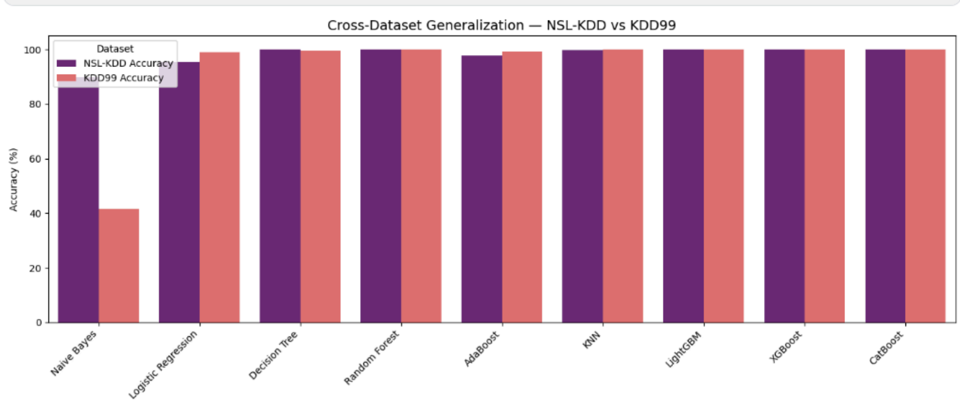
1. Checked and confirmed no missing values, no duplicate rows in NSL-KDD
2. Encoded protocol_type, service, flag using LabelEncoder — **fit only on train, transformed on test**, with an 'unknown' fallback category for values unseen during training (NSL-KDD's test set intentionally contains novel service/attack values)
3. Created binary target (Normal=0, Attack=1) and multiclass target (DoS, Normal, Probe, R2L, U2R)
4. Dropped the difficulty metadata column (not a real traffic feature) to prevent data leakage
5. Scaled all features with StandardScaler, fit on train only
6. For the KDD99 cross-dataset test: reused NSL-KDD's fitted encoders and scaler on KDD99 (no separate fitting), so both datasets are represented in the exact same numeric space

Machine Learning Models Implemented

Naive Bayes, Logistic Regression, Decision Tree, Random Forest, AdaBoost, KNN, LinearSVM, LightGBM, XGBoost, CatBoost — all trained and evaluated on NSL-KDD; the same trained models were then tested (not retrained) on KDD99.

Best Model Selected

- XGBoost best generalizer — 100% on KDD99
- Naive Bayes worst — 48% drop across datasets



Performance Metrics Achieved

NSL-KDD Binary Classification (Test Set)

Model	Accuracy	Precision	Recall	F1-Score
CatBoost	80.67%	—	—	80.08%
XGBoost	80.09%	96.78%	67.26%	79.36%
LightGBM	79.70%	96.76%	66.56%	78.87%
Decision Tree	78.85%	96.65%	65.11%	77.81%
AdaBoost	77.21%	96.29%	62.36%	75.70%
Random Forest	77.07%	96.62%	61.88%	75.44%
KNN	76.76%	97.12%	60.98%	74.92%
Naive Bayes	75.95%	96.48%	59.94%	73.94%
Logistic Regression	75.39%	92.53%	61.76%	74.08%
LinearSVM	75.15%	92.49%	61.33%	73.75%

Cross-Dataset Generalization (Trained on NSL-KDD → Tested on KDD99) — *provisional, rerun after your encoding fix before finalizing*

Model	NSL-KDD Accuracy	KDD99 Accuracy	Generalization Drop
Random Forest	99.99%	99.99%	0.00%
XGBoost	99.99%	100.00%	−0.01% (improved)
LightGBM	99.98%	99.99%	−0.01%
CatBoost	99.92%	99.99%	−0.07%
KNN	99.77%	99.95%	−0.18%
Decision Tree	99.99%	99.57%	0.42%
AdaBoost	97.69%	99.28%	−1.59%
Logistic Regression	95.40%	98.91%	−3.51%
Naive Bayes	89.93%	41.72%	**48.21%**

	Accuracy	Precision	Recall	F1 Score	Time (s)
XGBoost	99.89	99.48	99.86	99.67	3.03
LightGBM	99.88	99.39	99.87	99.63	5.12
CatBoost	99.86	99.37	99.81	99.59	42.59
Random Forest	99.79	99.43	99.32	99.38	45.11
Decision Tree	99.76	99.12	99.45	99.28	12.01
AdaBoost	99.13	98.58	96.22	97.39	46.98
KNN	98.97	96.89	96.99	96.94	39.23
Logistic Regression	95.56	88.56	84.59	86.53	7.40
Naive Bayes	79.23	42.13	61.95	50.15	0.25